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A review of the toxicological effects and allergenic potential of emerging alternative protein sources

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Abstract

The growing environmental pressure of the animal food chain requires a system shift toward more sustainable diets based on alternative protein sources. Emerging alternative protein sources, such as faba bean, mung bean, lentil, black gram, cowpea, quinoa, hemp, leaf proteins, microalgae, and duckweeds, are being explored for their potential in meeting global protein demand and were, therefore, the subject of this review. This systematic literature review aims to understand the current knowledge on the toxicological effects and allergenic potential associated with these sources and derived protein and food products. The findings identified potential concerns associated with the presence of plant secondary metabolites, including antinutritional factors, phytoestrogens, and oligosaccharides, in all the sources included. Also, these protein sources have been shown to display allergenic properties, either through the intrinsic presence of allergens or through cross-reaction. Further, the effects of food processing on these proteins remain poorly understood and no conclusive data are available to quantitatively assess their safety after processing. Overall, those findings highlight the need for quantitative knowledge of the food safety attributes related to final food products. This will enable a concrete and preventive approach to food safety in the protein transition.

KEYWORDS

allergenicity, alternative protein sources, antinutritional factors, food safety, microalgae

1 | INTRODUCTION

In the past decade, alternative protein sources for food have gained attention as substitutes for animal protein, praised for their favorable nutritional profiles and environmental sustainability (Eshel et al., 2019; Kim et al., 2019; Willett et al., 2019). The interest in more sustainable dietary patterns is prompted by the scarcity of natural resources, such as land and water, that underpin food production but have been overexploited in recent decades. This overexploitation challenges the whole ecosystem, making a turnaround of the agrifood system an imperative both to meet the growing demand for food and reverse the environmental pressure exerted to date. Currently, the global agrifood system is responsible for 34% of global anthropogenic greenhouse gas (GHG) emissions (Crippa et al., 2021; FAO, 2021). In detail, most of these originate from the production of foods of animal origin, that

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is, the traditional sources of protein in the Western diet (Poore & Nemecek, 2018; Sandström et al., 2018). In view of the above, the diversification of dietary protein sources is among the pillars of this transformation process to eventually enhance sustainability (Campbell et al., 2017; FAO & WHO, 2019). This shift involves incorporating a diverse range of alternative protein sources into the diet including; soil (or crop)-based sources such as legumes, seeds, leaves, and fungi, aquatic-based sources like macro- and microalgae or other aquatic plants, cell-based options, and proteins derived from insects. To ease the integration of alternative proteins in the diet, the global food industry is committed in the development of food products (named analogues) similar to those of animal origin but made from alternative protein sources, to expand the dietary choices of environmental-conscious consumers (Boukid, 2021). The first analogues introduced to the market were soy-based, as soybean has long been the prevalent alternative protein option (Rizzo & Baroni, 2018). However, the European "Farm-to-Fork" strategy advocates for a decrease in soybean usage in favor of a more diverse set of protein sources from EU countries (European Commission, 2020). Furthermore, soy remains a globally recognized allergens, which means strict food labeling rules are in place to protect allergic patients from experiencing adverse reactions (Gendel, 2012). In this regard, the food industry is moving toward the expansion of its production portfolio with the introduction of additional alternative protein sources; also in response to growing consumer demand (Banach et al., 2023; He et al., 2020). Nonetheless, novel protein sources must be carefully evaluated before entering the market since these might contain and/or expose consumers to a variety of food safety hazards, for example, chemical and microbiological contaminants, allergens, toxins, and secondary metabolites produced by plants. Safety aspects concerning emerging alternative protein sources and analogues are often largely unexplored or lacking sufficient knowledge (Banach et al., 2023; Milana et al., 2024; Mills et al., 2006; Thakur et al., 2019; Zhang et al., 2024). The present systematic review aims to obtain an up to date overview of the toxicological effects and allergic potential of alternative protein sources that are used or poised for use in the production of analogues. Specifically, the focus is on emerging protein sources of plant origin as these are readily available and accepted by consumers (Etter et al., 2024). Sources included were classified into soil-based (cowpea, faba beans, lentils, mung beans, black gram, quinoa, hempseeds, and leaf proteins) and aquatic-based (microalgae and duckweeds). The major goal of this review is to expand the understanding of the factors impacting the safety of alternative protein sources and, eventually support proactive efforts to protect the well-being of all consumers.

2 | METHODOLOGY

2.1 | Demarcation of the study

The section below presents a thorough description of the sources included. The research excluded well-established plant-based protein sources, which have been widely consumed in Europe for a long time and for which the toxicity and allergenic potential are known. Emerging alternative protein sources not of plant origin have been excluded as well.

2.2 | Alternative proteins of plant origin

2.2.1 | Soil-based

Faba beans (*Vicia faba*), mung beans (*Vigna radiata*), lentils (*Lens culinaris*), black gram (*Vigna mungo*), and cowpea (*Vigna unguiculata*) are members of the *Leguminosae* (or *Fabaceae*) family. Leguminous crops have multiple beneficial effects on the agroecological landscape owing to their contribution to biodiversity conservation and ability to resist drought and fix nitrogen. These characteristics support resource conservation and decrease the need for fertilizer and water (lower GHG emissions; FAO, 2016). From a protein-related nutritional perspective, legumes have good digestibility and protein content as well as balanced amino acid and essential amino acid (eAAs) profiles. As in most plant proteins, a slight decrease in sulfur-containing AAs is observed also in legumes (Affrifah et al., 2021; Boukid, 2024; Boukid & Castellari, 2022; FAO, 2016; Hertzler et al., 2020; Nasir & Sidhu, 2013; Nosworthy et al., 2017; Sudesh Jood, 1989; Torres-Tiji et al., 2020; Turck et al., 2021a).

In addition to legumes, quinoa (*Chenopodium quinoa*, *Amaranthaceae*) is another emerging alternative protein source. It is a gluten-free pseudocereal with both environmentally and nutritionally favorable characteristics, such as huge genetic variability, high-stress resistance, good protein content and digestibility, and balanced AAs profiles (eAAs in line with FAO/WHO requirements; Venlet et al., 2021; Vilcacundo Rubén, 2017).

Hemp (*Cannabis sativa*, *Cannabaceae*) is gaining interest as an alternative dietary source of plant proteins. It is a herbaceous annual plant whose environmental benefits include carbon storage and prevention of soil erosion (Yano & Fu, 2023). Although hemp has been traditionally grown for many uses (food, feed, textiles, etc.), since the late 1930s cultivation has been restricted in many countries due to the presence of delta-9-tetrahydrocannabinol (THC), a psychoactive substance, in the flowers. Nevertheless, varieties with THC content lower than 0.3% can be cultivated for industrial use (Yano & Fu, 2023).

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Nutritional characteristics include good protein digestibility and content and the presence of all eAAs (House et al., 2010).

Leaf proteins from *Medicago sativa* (or lucerne, alfalfa) and *Moringa* spp. (species *oleifera* and *peregrina*) plants are also regarded as emerging sources of plant proteins. From a nutritional perspective, the soluble fraction of leaves is mostly made from the protein Rubisco, which is known to contain all the eAAs for human requirements. Also, protein digestibility and quantity of both *M. sativa* and *Moringa* spp. are favorable (Anoop et al., 2023; Hadidi et al., 2023; Mielmann, 2013; Saa et al., 2019).

2.2.2 | Aquatic-based

Microalgae, which are a wide range of photosynthetic eukaryotic and cyanobacterial organisms, comprise millions of distinct species. While in Asia they are part of traditional culinary culture, in Western countries the interest has grown since mid-1900s for their content of beneficial compounds such as carotenoid pigments and omega-3 fatty acids (Enzing et al., 2014; Fu et al., 2021; Matos, 2019). The use of microalgae in food and supplements requires the approval of competent authorities. While *Chlorella* spp. are widely approved across continents, other species have been approved only in some countries. As a matter of fact, to date, *Chlorella* spp. is globally marketed as dietary supplement (Enzing et al., 2014; Moura et al., 2022; Torres-Tiji et al., 2020). In recent years, microalgae have also been recognized as a promising source of plant protein (Fu et al., 2021; Matos, 2019). From a protein-related nutritional perspective, microalgae are comparable to traditional animal and plant protein sources (Fu et al., 2021; Moura et al., 2022; Torres-Tiji et al., 2020). Microalgae are also considered a sustainable protein source as they can accumulate organic substances via solar energy and CO_2 and convert inorganic substances into valuable compounds (e.g., organic biomass; Matos, 2019). Duckweeds, or water lentils, are a diverse category of aquatic plants, primarily from the *Lemna* and *Wolffia* genera (Pyett et al., 2023). From a nutritional perspective, they contain a good and balanced amount of protein and eAAs; therefore, are currently investigated as an alternative source of protein for dietary uses (Hu et al., 2022; Mes et al., 2022; Xu et al., 2021). Duckweeds have a history of safe use in Asia while in Europe some species need "Novel Food" approval by the European Food Safety Authority (EFSA; EFSA, 2021; Pyett et al., 2023).

2.3 | Systematic literature search

Two categories of adverse human reactions were included, namely, toxicological effects and allergenic potential.

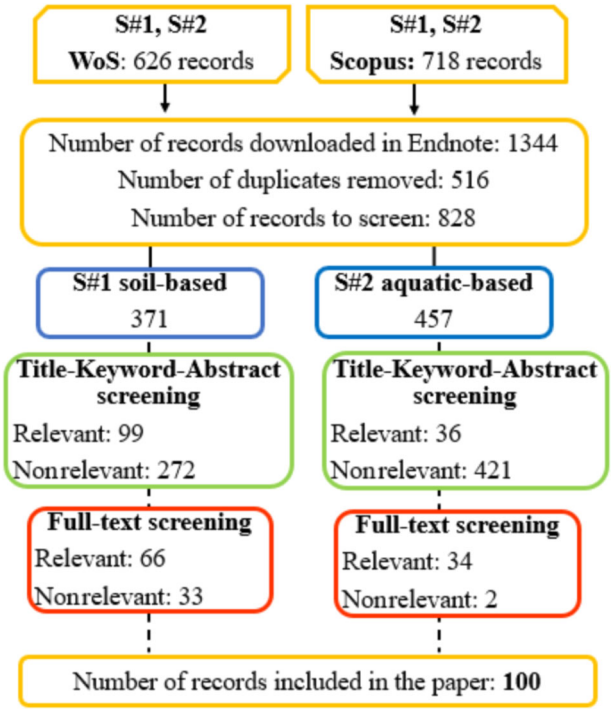

FIGURE 1 Flow diagram of the screening procedure (Microsoft PowerPoint).

Scopus and Web of Science (WoS) databases were utilized to conduct the literature searches, with a publication date from 2000 to 2023. The searches followed EFSA's guidelines for systematic reviews on food safety assessments as far as possible; each search query was defined through a combination of generic and specific keywords (EFSA, 2010; Table 1). After conducting the searches, the resulting references were downloaded into EndNote and duplicates were removed. The records from each search were screened by titles, keywords, and abstracts and, based on this evaluation, categorized as relevant and not relevant (Figure 1). To ensure consistency and reliability, a second reviewer independently screened 10% of the references for each search. Any discrepancies in categorization were discussed, and a consensus was achieved on inclusion and exclusion criteria. The papers included in the toxicological effects addressed acute, chronic, genotoxic, and carcinogenic effects as well as the presence of toxic compounds in the protein sources and derived foods. The papers included in the category of allergenic potential reported the presence of allergens, clinical reactions, and any cross-reactivities associated with the protein sources and derived foods. Studies focusing on beneficial effects such as anti-inflammatory potential and nutraceutical compound content were excluded from this review. Subsequently, all relevant and potentially relevant papers were read in full. Potentially relevant papers were judged

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TABLE 1 Search categories, search strings, and resulting searches defined to conduct the review.

Categories	
1. Generic soil-based protein* OR "plant based*" OR "plant protein based*" OR "plant protein*" OR "alternative protein*" OR "novel protein*" OR "meat substitute*" OR "meat analogue*" OR "meat alternative*" OR "novel food*"	
2. Generic aquatic-based protein* OR "microalgae*" OR "microalgae based food*" OR "aquatic protein*" OR "alternative protein*" OR "novel protein*" OR "novel food*"	
3. Food safety "food safety" OR food* OR safety* OR risk* OR "risk assessment*" OR health* OR exposure* OR outbreak* OR "human consumption*" OR "food safety risk" OR "food safety hazard*" OR "health concern*" OR "health risk"	
4. Specific soil-based protein sources pulse OR pulses OR legume* OR fa?a OR "vicia fa?a" OR "fa?a bean*" OR "broad bean*" OR "horse bean*" OR "field bean*" OR "mung bean*" OR "vigna radiata" OR "v. radiata" OR "green gram" OR cowpea* OR "black eye* pea*" OR "vigna unguiculata*" OR "v. unguiculata*" OR "crowder pea*" OR "southern pea*" OR lentil* OR "lens esculenta" OR "l. esculenta" OR "lens culinaris" OR "l. culinaris" OR "cicer lens" OR quinoa* OR "chenopodium quinoa" OR "c. quinoa" OR hemp* OR "lea? protein*" OR "green lea?* protein*" OR "lpc*" OR "edible lea?*" OR "medicago sativa" OR "m. sativa" OR alfalfa* OR lucerne* OR "cowpea lea?*" OR moringa* OR "moringa" oleifera lea?*	
5. Specific aquatic-based protein sources microalgae* OR cyanobacteria OR "green algae*" OR "edible algae" OR chlorella* OR "c. vulgaris" OR "c. pyrenoidosa" OR dunaliella* OR "d. salina" OR "d. tertiolecta" OR "aphanizomenon flos-aqua" OR "alga* fraction*" OR "alga* component*" OR duckweed* OR lemna* OR wolffia*	
6. Allergenicity **OR** Toxicity (allergen OR allergenicity OR "food allergy" OR "food allergen" OR "allergenic protein*" OR "allergic reaction" OR "cross-reaction" OR "cross-reactivity" OR "allergenic potential") OR (toxic* OR exposure* OR "*toxicity" OR dose* OR "reference value*" OR "adverse effect*" OR genotox* OR carcinogen*)	
Searches	
Soil-based	**Aquatic-based**
#1 Allergenic and Toxic potential 1. AND 3. AND 4. AND 6.	#2 Allergenic and Toxic potential 2. AND 3. AND 5. AND 6.

either relevant or not relevant after full-text reading and the findings of all relevant papers were summarized.

3 | RESULTS

The review procedure resulted in a total of 100 relevant papers, of which 61 on the toxicological effects related to alternative protein sources and 39 on the allergenic potential. Details of the distribution of papers per topic are given in Figure 2.

3.1 | Toxicological effects

The findings on toxicological effects mainly concern the presence of plant secondary metabolites, such as antinutritional factors (ANFs), phytoestrogens, and oligosaccharides, as well as the outcomes of toxicological studies, targeting the protein sources and some derived protein ingredients. ANFs are produced as a defense mechanism against biotic and abiotic stressors from insects, microorganisms, and also environmental conditions. From a nutritional perspective, these are considered antinutrients as they lower the quality of plant proteins by interfering with the bioavailability and absorption of nutrients. Phytoestrogens can either have therapeutic or adverse effects, as they act as endocrine disruptors and can alter the functioning of different body systems leading to, for example, impaired fertility, growth inhibition, and altered food intake. Similarly, oligosaccharides are short-chain sugars not fully digested by humans which may act as antinutrients by causing digestive discomfort (Awan et al., 2014; Sridevi et al., 2021). Table 2 shows a summary of the findings from this section. Detailed findings are described below.

3.1.1 | Soil-based alternative protein sources

The content of ANFs in legumes has been broadly investigated. For lentils, findings reported the following concentrations. Total phenols: 6.56 mg GAE/g (gallic acid equivalent) with 9.15 mg/g of tannins and 5.97 mg cat eq/g (catechin equivalent) of condensed tannins, phytic acid: 12.50 mg/g, saponins: 3.50 mg/g, lectins: 1 mg/g and hemagglutination activity of 2.7×10^{-3} mg/g (Ahuja et al.,

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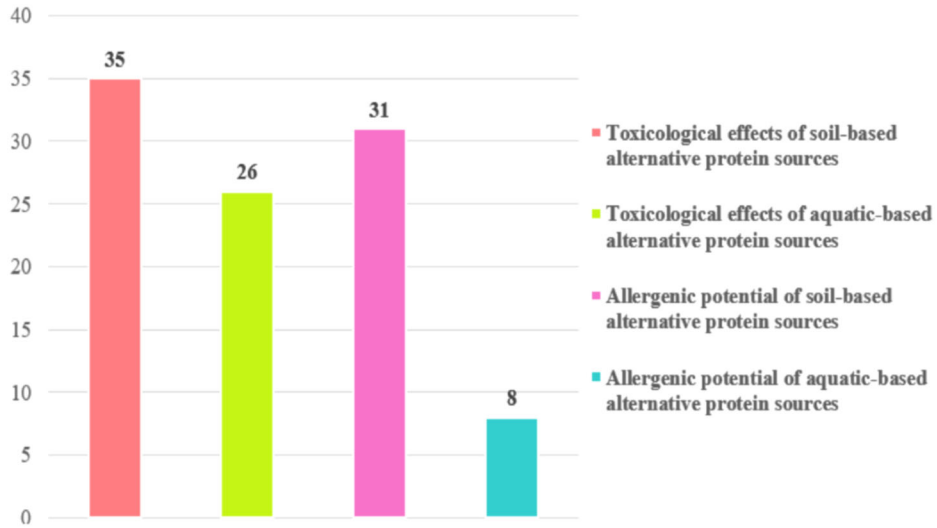

FIGURE 2 Distribution of the papers included classified per topic and group of alternative protein sources (Microsoft PowerPoint).

2015; Awan et al., 2014; Martin-Cabrejas et al., 2009; Nasi et al., 2009; Zia ur & Salariya, 2005). In lentils, protease inhibitory activity can inhibit trypsin and chymotrypsin simultaneously at independent binding sites; the highest trypsin inhibitory activity has been reported as 65.60 mg/g (Ahuja et al., 2015; Awan et al., 2014). EFSA reported the content of 2.80 mg/g of L-canavanine, a nonprotein amino acid, in lentil flour (EFSA, 2009). Lentils also contain the following phytoestrogens; isoflavones (> 0.1 mg/100 g) and flavonoids (1.30 mg cat eq/g) with 3.24 mg/g of catechins (Awan et al., 2014; Martin-Cabrejas et al., 2009; Radd, 2000; Sridevi et al., 2021). Further, oligosaccharides such as raffinose (0.42 dry weight [dw]), stachyose (1.87 dw), and verbascose (0.49 dw) are also found in lentils (Wang & Daun, 2006; Table 3). Faba beans are known for causing hemolytic anemia (favism) in susceptible individuals (i.e., those with a deficit in the glucose-6-phosphate dehydrogenase [G6PD] enzyme). This is due to the presence of pyrimidine glycosides (i.e., vicine and convicine), which are responsible for glutathione depletion and high molecular weight protein cluster formation. The content of pyrimidine glycosides in faba bean is up to 7.01 mg/g (vicine) and 3.12 mg/g (convicine; Agrawal et al., 2024; Mayer Labba et al., 2021). Overall, the following concentrations of ANFs have been reported in faba beans. Total phenols: 5 mg GAE/g with 6.5 mg/g dw of tannins and 1.95 g eq cat/KT of condensed tannins, phytic acid: up to 21,700 mg/g, saponins: 4 mg/g, and lectins: 1 mg/g (Alonso et al., 2000; Mayer Labba et al., 2021; Nasi et al., 2009; Sharma, 2021). Concerning enzyme inhibitors, trypsin and chymotrypsin inhibitor activities have been reported as 4.47×10^{-3} and 3.65×10^{-3} IU/g dw, respectively, α-amylase inhibitor activity as 18.9 IU/g dw and hemagglutinating activity as 4.93×10^{-2} HU/g dw (Alonso et al., 2000; Mayer Labba et al., 2021; Sharma, 2021). Faba beans also contain phytoestrogens such as isoflavones (0.1 mg/100 g) and oligosaccharides such as raffinose (up to 4 mg/g dw), stachyose (16 mg/g dw), and verbascose (34 mg/g dw; Radd, 2000; Sharma, 2021; Table 3). In cowpeas, the hemagglutinating activity is high in the albumin fraction (up to 444.40 HU/g in flour) while the content of lectin is relatively low (3×10^{-8} HU/g; Awan et al., 2014). Findings on trypsin inhibitor activity reported were 12.4 kIU/g dw in cowpeas and 8×10^{-3} IU/g sample in whole bean cowpea flour (Frota et al., 2018; Sharma, 2021). Further, Frota et al. (2018) reported that trypsin inhibitory activity in protein isolates was 1.06×10^{-2} IU/g sample (heat-treated isolates) and 7.8 IU/mg sample (nonheat-treated isolates) and Dilrukshi et al. (2022) reported an increase in trypsin inhibitory activity (up to 23.83 IU/mg) and phytic acid content (up to 365.05 mg/100 g) with increasing cowpea fraction in different flour formulations. Cowpea also contains phytoestrogens such as coumestans and oligosaccharides such as raffinose (8 mg/g dw), stachyose (52 mg/g dw), and verbascose (8 mg/g dw; Sharma, 2021; Sridevi et al., 2021; Table 3). According to Awan et al. (2014), mung bean consumption can produce a small increase in the blood glycemic index in humans, making it an unfavorable option for diabetic consumers. The content of ANFs has been reported for different mung bean protein ingredients. In mung bean flour, concentrations found were as follows: tannin (8 mg/g), phytic acid (> 8 mg/g), saponin (> 10 mg/g), trypsin and α-amylase inhibitors (96.8 and 3.68 IU/g, respectively), and hydrogen cyanide (34.60 mg/g). The flour also contains oligosaccharides of the raffinose series, that is, 10.66 mg/g (bound fructose; Batra et al., 2019). In protein extracts, concentrations found were as follows: lectins (120 HU/g),

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TABLE 2 Summary of the findings on the toxicological effects of the alternative protein sources included in this review.

	Black gram	Cowpea	Faba bean	Lentils	Mung bean	Quinoa	Hemp	Leaf proteins	Microalgae	Duckweeds
Secondary metabolites										
Alkaloids									x	
Biogenic amines	x									
Chlorophylls									x	
Chymotrypsin inhibitors			x	x		x				
Hemagglutination activity	x	x	x	x					x	
Hydrogen cyanide					x					
Lectins		x	x	x	x				x	
L-canavanine				x				x		
Oxalates						x				x
Oxalic acid										x
Oligosaccharides	x	x	x	x	x					
Phenolic compounds	x		x	x	x	x	x		x	x
Phytates							x			
Phytic acid	x	x	x	x	x	x	x		x	x
Phytoestrogens		x	x	x		x		x	x	x
Polyphenols				x	x	x	x			
Pyrimidine glycosides			x							
Saponins	x		x	x	x	x		x	x	
Trypsin inhibitors	x	x	x	x	x	x	x		x	x
α-Amylase inhibitors			x		x					
Adverse health effects										
Accumulation and transfer of POPs									x	
Electrolyte imbalance								x		
Favism			x							
Increased glucose level/glycemic index					x			x		
No observed adverse effect level (NOAEL) values								x	x	x
Potential adverse effects on the bone marrow								x		
Potential adverse effects on the heart								x		
Potential kidney damage								x		
Potential hepatic damage								x		
Potential polythemia								x		
Potential cytotoxicity to mammalian cells									x	
Systematic lupus erythematosus activation								x		

phytic acid (13.30 mg/g), trypsin inhibitors (5.88×10^{-3} IU/g), and condensed tannins (8.59 µg/g dw; Turck et al., 2021a). Regardless of ANFs found in mung bean, in vivo toxicity of texturized mung bean proteins resulted in no signs of adverse effects (i.e., plasma biochemical analysis, organ weights, and histological characteristics) in rats (Brishti et al., 2020). Further, mung beans also contain phytoestrogens such as lignans; specifically 4×10^{-3} mg/g of secoisolariciresinol has been reported (Hussain Zaki et al., 2023; Table 3). Black gram is reported to contain

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TABLE 3 Occurrence and concentrations of plant secondary metabolites in the soil-based alternative protein sources as presented in the papers consulted (mg/g unless otherwise specified).

	Black gram	Cowpea	Faba bean	Lentils	Mung beans	Quinoa	Hemp	*Medicago sativa* (c)
Biogenic amines	0.30							
Chymotrypsin inhibitors			3.65×10^{-3} IU/g	✓		✓		
Flavonoids *Catechins*				1.30 cat eq 3.24				
Hemagglutination activity	0.20 kU/g	444.40 HU/g (f)	4.93×10^{-2} HU/g	2.7×10^{-3}				
Hydrogen cyanide					34.60 (f)			
Lectins		3×10^{-8} HU/g	1000	1000	120 HU/g (e)			
Lignans *Secoisolariciresinol* *Lignanamides*					4×10^{-3}	✓	0.77	
L-canavanine				2.80 (f)				4.3×10^{-3}
Oligosaccharides *Raffinose* *Stachyose* *Verbascose*	1.26 dw 5.40 dw 31 dw	8 dw 52 dw 8 dw	4 dw 16 dw 34 dw	0.42 dw 1.87 dw 049 dw	✓(f)			
Oxalates						2.32		
Phenolic compounds *Total phenols* *Condensed tannins* *Tannins*	8.90		5 GAE eq 1.95 g cat eq/KT 6.5	6.56 GAE eq 5.97 cat eq 9.15	8.59×10^{-3} (e) ✓ (f)	5.97 GAE/g 0.31	2.21 GAE/g 1.05	
Phytates							31	
Phytic acid	11	✓	21,700	12.50	✓ (f), 13.30 (e)	12	35	
Phytoestrogens *Coumestans* *Coumestrol* *Isoflavones* *Flavonoids*		✓	0.1 mg/100 g	>0.1 mg/100 g 1.30 cat eq		✓		7.8×10^{-2} 2.55×10^{-1}
Pyrimidine glycosides *Vicine* *Convicine*			7.01 3.12					
Saponins	2300		4	3.50	✓ (f)	5.57		1.4%

(Continues)

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TABLE 3 (Continued)

	Black gram	Cowpea	Faba bean	Lentils	Mung beans	Quinoa	Hemp	*Medicago sativa* (c)
Trypsin inhibitors	120 kU/g	12.4 kU/g 8×10^{-3} IU/g* (f) 7.8 IU/mg* (i)	4.47×10^{-3} IU/g	65.60	96.80 IU/g (f) 5.88×10^{-3} IU/g (e)	5.04×10^{-3} IU/g	✓	
α-Amylase inhibitors			18.9 IU/g		3.68 IU/g (f)			

c, protein concentrate; e, protein extract; i, protein isolate; f, flour; dw, dry weight; ✓, qualitative knowledge; U, units; IU, inhibitory units; cat eq, catechin equivalent; HU, hemagglutination units; GAE, gallic acid equivalent; *, measurement are per sample.References: Ahuja et al. (2015), Alonso et al. (2000), Awan et al. (2014), Batra et al. (2019), Bessaire et al. (2021), Dilrukshi et al. (2022), EFSA (2009), Escobar-Saez et al. (2022), Frota et al. (2018), Martin-Cabrejas et al. (2009), Mattila et al. (2018), Mayer Labba et al. (2021), Nasi et al. (2009), Pathan and Siddiqui (2022), Radd (2000), Sharma (2021), Sridevi et al. (2021), Turck et al. (2021a), Wang and Daun (2006), Hussain Zaki et al. (2023), Zia ur and Salariya (2005).

the following ANFs: tannins up to 8.90 mg/g, phytic acid up to 11 mg/g, biogenic amines at 0.30 mg/g, saponins at 2300 mg/g, trypsin inhibitor activity at 120 kU/g, and hemagglutinating activity at 0.20 kU/g dw (Awan et al., 2014; Sharma, 2021; Zia ur & Salariya, 2005). Black gram also contains oligosaccharides at concentrations of 1.26 mg/g dw (raffinose), 5.40 mg/g dw (stachyose), and 31 mg/g dw (verbascose; Sharma, 2021; Table 3).

Apart from legumes, ANFs and phytoestrogens were also reported in leaf proteins. Concerning a protein concentrate from alfalfa (APC), the concentrations are as follows: saponins (1.4%), L-canavanine (4.3×10^{-3} mg/g), coumestrol (7.8×10^{-2} mg/g), total polyphenols (20 mg/g), and isoflavones (average: 2.55×10^{-1} mg/g; with daidzein: 2.4×10^{-2} mg/g, genistein: 1.01×10^{-1} mg/g, and glycitein: 1.30×10^{-1} mg/g; EFSA, 2009; Table 3). No adverse effects were shown after consumption of a single dose of APC of 5000 mg/kg body weight (bw). Besides, the consumption of alfalfa-based dietary supplements (from seeds and sprouts) has been associated with the occurrence of systemic lupus erythematosus activation, which is linked to the high content of L-canavanine in the plants (80–150 mg/kg in seeds and 10 mg/kg in leaves; EFSA, 2009). More findings were retrieved on the effects of *Moringa* spp. consumption. Acute toxicity tests of different plant parts, extracts, and derived products (leaves infusion and powder) of *M. peregrina* and *M. oleifera* showed no clinical signs of toxicity in rats at an oral dose of 2000 mg/kg bw (Adedapo et al., 2009; de Barros et al., 2022; El-Hak et al., 2018; Saleem et al., 2020). Despite this, de Barros et al. (2022) reported that leaves infusion and powder administration above 2000 mg/kg bw, and powder at 500 and 1000 mg/kg might predispose to hepatic and kidney damage (increase in serum liver enzymes, i.e., alanine transaminase or ALT, aspartate aminotransferase or AST, urea nitrogen, and creatinine) during chronic administration (Oyagbemi et al., 2013). As a result, the no observed adverse effect level (NOAEL) has been set at 1000 mg/kg for infusion and 250 mg/kg for powder (de Barros et al., 2022). Additional acute toxicity tests of a *M. oleifera* alcoholic extract at oral doses ranging from 1 to 20 g/kg bw produced dose–responses mortalities among rats (LD_{50}: 4.0 g/kg bw; Chivapat et al., 2012; Igbokwe et al., 2018). Chronic toxicity tests of *M. oleifera* alcoholic extracts have also been performed. Igbokwe et al. (2018) reported that oral doses of 100 and 500 mg/kg bw for 30 days produced the following effects: (1) electrolyte imbalance; (2) increased production of blood components (potential *M. oleifera*-associated polythemia), liver enzymes ALT and AST (potential adverse effects on the liver), total protein, urea, bilirubin, and creatinine; (3) decreased volume of white blood cells and mean corpuscular volume (potential adverse effects on the bone marrow) and liver enzymes

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AST and alkaline phosphatase or ALP (potential adverse effects on heart); and (4) edematous hepatic tissue at low and median concentrations (absent at high) and diffused renal corpuscles. Chivapat et al. (2012) reported the following effects in rats that were given oral doses of 100, 500, and 1000 mg/kg bw for 90 days: (1) red blood and eosinophil cells decrease, (2) total protein decrease, (3) glucose levels increase, and (4) no lesions to visceral organs, with minor differences among male and female rats. The protein fraction of *M. oleifera* is considered noncytotoxic as in silico assessment of the peptides found no immunogenicity, carcinogenicity, or mutagenicity (Atolani et al., 2021).

Hemp seeds are also found to contain the following ANFs: 1.05 mg/g dw of condensed tannins, 35 mg/g dw of phytic acid and up to 31 mg/g of phytates (Escobar-Saez et al., 2022; Mattila et al., 2018). Mattila et al. (2018) reported the content of total phenolic compounds as 0.96 mg GAE/g or 2.21 mg GAE/g depending on the analytical method used. Further, the content of lignanamides (a class of phenolic compounds) in hemp seeds was reported at 0.77 mg/g dw. Hemp seeds also showed strong trypsin inhibitory activity (Mattila et al., 2018; Table 3).

For quinoa, the following concentrations of ANFs were reported: Phytic acid: 12 mg/g dw, total saponins: 5.57 mg/g dw, oxalates: 2.32 mg/g, and tannins: 0.31 mg/g (Mattila et al., 2018; Pathan & Siddiqui, 2022). The content of total phenolic compounds has been reported as 1.81 mg GAE/g or 5.97 mg GAE/g depending on the analytical method used (Mattila et al., 2018). Quinoa also showed mild trypsin and chymotrypsin inhibitory activity, with the highest trypsin inhibitor content of 5.04×10^{-3} IU/g (Mattila et al., 2018; Pathan & Siddiqui, 2022). Further, quinoa contains phytoestrogens such as lignans and the isoflavones genistein and genistin (Bessaire et al., 2021; Radd, 2000; Sridevi et al., 2021; Table 3).

Overall, several studies indicated that different processing methods, such as using high temperatures, fermentative microorganisms, or other physicochemical treatments, can significantly lower the content of ANFs in the protein sources and protein extracts (i.e., isolates, concentrates, and texturized products) examined in this review (Abd El-Hady & Habiba, 2003; Batra et al., 2019; Brishti et al., 2020; Dilrukshi et al., 2022; Hall & Moraru, 2021b; Martin-Cabrejas et al., 2009; Nasi et al., 2009; Sharma, 2021; Zia ur & Salariya, 2005). However, quantitative data on processing effects are lacking.

3.1.2 | Aquatic-based alternative protein sources

For duckweeds, the searches found multiple EFSA reports following Novel Food applications for *Lemna* and *Wolffia* spp. either as whole plants or derived products (powder and protein concentrate). The reports indicated no signs of toxicity after administration of oral doses of *Lemna* spp. powder at 1000 mg/kg bw in rats. For *W. globosa* powder, a NOAEL of 6.5 g/kg bw/day was established. Overall, no toxicological concerns about genotoxicity have emerged (Turck et al., 2021b, 2023). Concerning the content of secondary metabolites, the reports indicated oxalic acid activity in duckweed-based food products while tannins, phytic acids, and trypsin inhibitors have only been reported in *Lemna* species powder and protein concentrate (Turck et al., 2021b, 2022, 2023). Additional findings on *W. arrhiza* reported the presence of oxalates (8.04 mg/g dw), phytic acid (0.22 mg/g dw), phenolic compounds (7.57 mg/g dw) with 9.83 mg cat eq/g dw of tannins, and 41.20 mg/g dw of flavonoids (Hu et al., 2022; Table 4).

For microalgae the content of ANFs and phytoestrogens have been reported in different species (Ridlo et al., 2023). Alkaloids, saponins, steroids, and flavonoids have been identified in *Dunaliella salina* and *Botryococcus braunii* extracts. The total phenolic content of the microalgae extracts was 61.28 mg GAE/g sample for *D. salina* and 46.94 mg GAE/g sample for *B. braunii*. *D. salina* contain chlorophyll a and chlorophyll b at 11.11 and 7.25 mg/g, respectively, while *B. braunii* at 18.15 and 12.59 mg/g, respectively (Ridlo et al., 2023). Findings on *Chlorella pyrenoidosa* reported the content of saponins: 40.68 mg/g (57.50 mg/g in *Chlorella vulgaris*), phytic acid: 0.62 mg/g, tannins: 2.66 mg/g, trypsin inhibitor activity (TIA): 0.52 mg/g, and lectins (Chen et al., 2022). Further, the content of lectins and hemagglutination activity have been identified in different microalgae such as *Ankistrodesmus fusiformis*, *Chlamydocapsa bacillus*, *Coelastrum microporum*, *Desmodesmus brasiliensis*, *Kirchneriella lunaris*, *Pseudokirchneriella subcapitata*, *Scenedesmus obliquus*, and *Tetradesmus obliquus* (De Carvalho Carneiro et al., 2019; Silva et al., 2020; Table 4). Apart from secondary metabolites, findings on the toxicological effects associated with microalgae are reported below. Focusing on *Chlorella* spp., our searches found different studies assessing its toxicity. Day et al. (2009) determined the NOAEL values for male and female rats in a 28-day study with *C. protothecoides* biomass. The estimated values are 7557 mg/kg bw/day (males) and 8068 mg/kg bw/day (female). Further, tests on the subchronic toxicity and genotoxic potential of protein (p) and flour (f) preparations from *C. protothecoides* indicated the following NOAEL values: 4805 (p) and 4807 (f) mg/kg bw/day for males, and 5518 (p) and 5366 (f) mg/kg bw/day for females (Szabo et al., 2012, 2013). Similar findings were attributed to *C. sorokiniana*; NOAEL in male and female rats were determined at 5940 mg/kg bw/day and 6410 mg/kg bw/day, respectively (Himuro et al., 2017). De Mello-Sampayo et al. (2013)

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TABLE 4 Occurrences and concentrations of plant secondary metabolites in the aquatic-based alternative protein sources (mg/g unless otherwise specified).

	Duckweeds	Microalgae
Alkaloids		✓ *(D.) (B.)*
Chlorophylls *Chlorophyll-a* *Chlorophyll-b*		11.11 *(D.)*, 18.15 *(B.)* 7.25 *(D.)*, 12.59 *(B.)*
Flavonoids	41.20	✓ *(D.) (B.)*
Hemagglutination activity		✓*
Lectins		✓ *(C.)* *
Oxalates Oxalic acid	8.04 *(W.)* ✓ (duckweed-based food)	
Phenolic compounds *Total phenols* *Condensed tannins* *Tannins*	7.57 *(W.)* 9.83 cat eq *(W.)*, ✓ *(L.)* (c)	61.28 GAE *(D.)*, 46.94 GAE *(B.)* 2.66 *(C.)*
Phytic acid	0.22 *(W.)*, ✓ *(L.)* (c)	0.62 *(C.)*
Saponins		✓ *(D.) (B.)*, 57.50 *(C.)*
Terpenoids *Steroids* *Triterpenoids*		✓ *(D.) (B.)* ✓ *(D.)*
Trypsin inhibitors	✓ *(L.)* (c)	0.52 *(C.)*

L., *Lemna* spp.; c, protein concentrate; eq cat, catechin equivalent; GAE, gallic acid equivalent; ✓, qualitative knowledge; W., *Wolffia* spp.; D., *Dunaliella salina*; B., *Botryococcus braunii*; C., *Chlorella* spp.; *, *Ankistrodesmus fusiformis*, *Chlamydocapsa bacillus*, *Coelastrum microporum*, *Desmodesmus brasiliensis*, *Kirchneriella lunaris*, *Pseudokirchneriella subcapitata*, *Scenedesmus obliquus*, and *Tetradesmus obliquus*.References: De Carvalho Carneiro et al. (2019), Chen et al. (2022), Hu et al. (2022), Ridlo et al. (2023), Silva et al. (2020), Turck et al. (2021b, 2022, 2023).

reported no signs of toxicity either after a single administration of 125 mg/kg bw or after repeated administrations in a stress-induced carotenogenic biomass of *C. vulgaris*. Another study reported that *Chlorella* and *Scenedesmus* spp. growth under iron depletion (common during algal biomass production) resulted in solvent-fractionated samples, which exhibit cytotoxicity when presented to mammalian cells (Bagwell et al., 2016). *Chlorella* spp. can also accumulate hexachlorobenzene (HCB; a toxic persistent organic pollutant) and transfer it to crabs (*Chasmagnathus granulatus*). This leads to toxic effects in both organisms (i.e., decreased uroporphyrinogen decarboxylase [UROD] activity, induced lipid peroxidation, decreased antioxidant defenses, and disorganization of the epithelium in hepatopancreas tubules; Chaufan et al., 2006). Interestingly, Day et al. (2009) pointed out that due to broad similarities among the different species of *Chlorella*, it is possible to consider the results of toxicity tests as applicable to all. Beyond *Chlorella* spp., no signs of toxicity were reported after a 3-month administration of 100 mg/kg bw/day of a powder from *D. salina* (El-Baz et al., 2019). Similarly, the β-carotenoid-rich fraction of *D. salina* leads to no acute and chronic toxic effects at administration doses of 5000 mg/kg bw/day and 500 mg/kg bw/day, respectively (El-Baz et al., 2022b). Three *D. salina* extracts (methanol, ethyl acetate, and *n*-hexane) were evaluated for their content of phytochemicals and toxicity; alkaloids, steroids, triterpenoids, and phenols were detected in all, saponins in the methanol extract only, and flavonoids in both the methanol and ethyl acetate extracts. The following lethal toxicity values for the extracts were reported (lethal concentration for 50% of tests; LC_{50}): 276 ppm (*n*-hexane), 811 ppm (methanol), and 673 ppm (ethyl acetate; Sijabat et al., 2023). Apart from *D. salina*, a NOAEL of 4000 mg/kg bw/day was estimated for the biomass of *Chlamydomonas reinhardtii* (Murbach et al., 2018). The toxicity of the astaxanthin-rich fraction of microalgae *Haematococcus pluvialis* has been tested in rats and mice. Overall, no adverse effects were found in experimental animals; Stewart et al. (2008) reported NOAEL values of 14,161 mg/kg bw/day (males) and 17,076 (females) mg/kg bw/day while El-Baz et al. (2022a) reported no adverse effects at a dose of 500 mg/kg of chronic administration. The toxicity of DHA-microalgae *Schizochytrium* spp. has also been tested in growing swine, rats, and rabbits. In commercial strains of swine, it did not produce any treatment-related toxic effects. In rats, the no observed effect level (NOEL) for maternal and developmental toxicity was determined at 22 g/kg a day while the absence of toxic effects has been determined at 4000 mg/kg a day of chronic exposure. In rabbits, the NOEL for maternal toxicity was determined at 600 mg/kg a day and at 1800 mg/kg a day for developmental toxicity

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(Abril et al., 2003; Hammond et al., 2001a, 2001b, 2001c, 2002).

3.2 | Allergenic potential

The findings on allergenic potential mainly concern the presence of allergens, the existence of cross-reactivities and the history of sensitization related to the different protein sources. It is important to consider these three factors when assessing the allergenic potential of a food, particularly when dealing with novel food sources. Beyond the intrinsic presence of allergens or allergen-like proteins, the history of sensitization of the food, as well as phylogenetic relationships and structural similarities with known allergens (which might result in cross-reactions), must be carefully considered. In detail, sensitive individuals can manifest allergic reactions either through direct sensitization to an allergen or through cross-reactivity (Cox et al., 2021). Table 5 shows a summary of the findings according to the three factors described above and Table 6 provides details on the dietary-related (established and potential) cross-reactivities.

3.2.1 | Soil-based alternative protein sources

Legumes are generally considered common allergenic foods as their intrinsic allergens have been widely reported and described (Verma et al., 2012, 2013). For lentils, the three major allergens are named Len c1, Len c2, and Len c3 (Cabanillas et al., 2018; Halima et al., 2022; Shaheen et al., 2019). Len c1 is a storage protein from the cupin superfamily, whose proteins L1, L2, L3 were first identified by Sánchez-Monge et al. (2000). The genetic variant Len c1.01 (a 48 kDa vicilin) was later identified as a major lentil allergen; it can inhibit up to 65% of IgE-binding to crude lentils and induce immunological reactions in 77% of individual lentil-allergic patients sera. Furthermore, two Len c1.01 fragments of 12–16 kDa (Len c 1.02 or Lenc c1) and 26 kDa can also inhibit IgE-binding. The amino acid sequences of Len c1.01 and its fragments displayed sequence homology with other vicilins, that is, allergens from peanut (Ara h 1), soybean (conglutinin subunit), lupine (β-conglutins), walnut (Jug r2), cashew (Ana o1), and sesame (Ses i3; Koeberl et al., 2018; Lopez-Torrejon et al., 2003). The other major allergens are a seed biotinylated protein (Len c2) and a nonspecific lipid transfer protein (ns-LTPs) from the prolamine superfamily (Len c3; Akkerdaas et al., 2012; Sánchez-Monge et al., 2000). The LTPs are considered pan-allergens as they are widespread in the plant kingdom; as it stands six Len c3 isoforms are known (Halima et al., 2022). The Len c3 showed specific IgE (sIgE) in sera from lentil-allergic patients and a complete inhibition by other LTPs, such as the peach allergen Pru p3 (Akkerdaas et al., 2012). In addition, recent studies applied in silico analyses to further explore lentil allergenicity. The proteomic profiling of lentil's low molecular weight proteins identified 44 potential allergens and allergen-like proteins (Shaheen et al., 2019); while other analyses performed via AllerCatPro (AllerCatPro 2.0 [a-star.edu.sg]; a web tool for predicting protein allergenicity potential) identified 22 strong allergens and 19 weak allergens (Halima et al., 2022). In detail, the web tool identified two pathogenesis-related proteins (PRs), that is, PR4 and PR10b, and a disease resistance response protein (DRR), that is, DRR49, as potential novel lentil allergens. Also, PR10b and DRR49 showed sequence similarity with known allergens and evolutionary analyses proved epitope conservation and evolutionary relation with PRs and DRRs proteins from other legume species (i.e., mung bean, soybean, lupin, and peanut; Halima et al., 2022). Overall, the presence of additional lentil allergens (i.e., provicilin, legumin A2, and LTPs) is highly variable across different samples and varieties (Shaheen et al., 2019). Concerning lentil cross-reactivity that with lupine and other legumes such as chickpeas and peas are extensively reported in the literature (Cabanillas et al., 2018; Cuadrado & Pedrosa, 2017; Koeberl et al., 2018; Reche et al., 2013). In detail, Cuadrado and Pedrosa (2017) reported 40% sequence homology and 70% sequence similarity among lupin (Lup-1) and lentil (Len c1.01) allergens. Moreover, clinical and immunological adverse reactions to lupine have been reported in lentil-allergic adults (Cabanillas et al., 2018). In addition, cross-reactivity between lentils and coconut have been found as Manso et al. (2010) reported the occurrence of an allergic reaction in a lentil-allergic child after consumption of prawns cooked with coconut milk. Skin prick tests (SPTs) for prawns and other major food allergens were all negative, except for those testing with lentils and coconut. Also, the oral food challenge with prawns produced no reaction. A cross-reactivity between lentils and coconut was later confirmed for the first time by the IgE inhibition assay. For faba beans, eight proteins (molecular weight: 70, 60, 48, 32, 23, 19, 15, 10 kDa) have been identified as potential allergens as showing allergen-like characteristics (i.e., resistance to digestion) and the ability to induce IgE-mediated allergic responses in sensitized mice and allergic patients (Kumar et al., 2014). Furthermore, cross-reactivities were identified, by ELISA tests, with peanut, fenugreek, red kidney bean, red gram, green gram, chickpea, and black gram (Kumar et al., 2014). Also, Cho et al. (2017) reported that soybean can be cross-reactive with faba bean, Koeberl et al. (2018) reported that lupine can be cross-reactive with faba bean and Jensen et al. (2015) that peanut-allergic patients gave positive SPTs and immunological responses

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TABLE 5 Summary of the findings on the allergenic potential of the alternative protein sources included in this review.

	Black gram	Cowpea	Faba bean	Lentils	Mung bean	Quinoa	Hemp	Leaf proteins	*Chlorella* spp.	Duckweeds
Intrinsic presence of allergens										
Allergens classified by WHO/IUIS				x	x		x(a)			
Allergens not classified by WHO/IUIS	x	x	x	x					x	
Potential allergens identified in silico				x				x	x	
Potential of cross-reactivities										
Cross-reactivity with nondietary allergens	x				x		x			
Cross-reactivity with dietary allergens or foods	x		x	x	x	x	x (p)	x		
Phylogenetic relation with known allergens				x	x					
Sequence similarities with allergens or proteins	x	x		x	x		x	x	x	x
History of sensitizations										
History of sensitization to allergens or food	x	x	x	x	x		x (nd)		x (s)	
Generation of positive immunological responses		x	x	x	x			x	x (s)	
Generation of immunological response to other allergens or foods		x	x	x	x			x		
Negative sensitization test to derived food products									x	
Effect of processing										
Allergenicity can be reduced via enzymatic hydrolysis				x						
Allergenicity can be reduced via fermentation				x						
Allergenicity can be reduced via protein isolation							x			
Allergenicity can be reduced via thermal treatments	x			*						

p, potential; a, airway allergens; nd, nondietary; s, food supplement; *, only autoclaving can reduce it; WHO/IUIS, World Health Organization and International Union of Immunological Societies Allergen Nomenclature Subcommittee.

to faba bean. For cowpea, four allergens have been identified in cowpea as part of cupin (vicilin) and prolamin (seed albumin proteins) superfamilies. Cowpea sensitization was tested in legumes-allergic children through SPTs, serum sIgE, IgE-inhibition assay (for cross-reactivity with other legumes) and basophil activation test (BAT; for sensitivity and reactivity toward legumes). As a result, most of the children were sensitized to cowpeas with a higher prevalence among the peanut-allergic ones. Also, cowpea extracts strongly inhibited IgE-binding to pea and peanut ones, plus sequence similarity was reported among cowpea allergens and allergens of peanut, pea, lentil, soybean, mung bean, and lupine (Chentouh et al., 2022). For mung beans, six allergenic sequences have been identified. These were named as follows: Vig r1 (pathogenesis-related protein, 16 kDa), Vig r2 (52 kDa, vicilin-type seed storage protein), Vig r3 (renamed Vig r2.0201, 50 kDa, vicilin-type seed storage protein), Vig r4 (30 kDa, seed albumin protein), Vig r5 (recently identified as a fragment of Vig r2, 18 kDa), and Vig r6 (cytokinin-specific binding protein, 18 kDa; WHO/International Union of Immunological Societies [IUIS] Allergen Nomenclature Subcommitee). The allergens Vig r2, Vig r3, Vig r4, and Vig r5 (identified in mung bean seeds) showed allergen-like characteristics

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TABLE 6 Potential and established dietary cross-reactivities of the alternative protein sources included in this review.

	Black gram	Cowpea	Faba bean	Lentils	Mung bean	Quinoa	Hemp	Leaf proteins	*Chlorella* spp.	Duckweeds
Apples					s			s		
Atlantic salmon									s	
Black gram			c							
Cashew				s						
Celery										s
Chickpea			c	s						
Chinese white shrimp									s	
Coconut				c,i						
Corn								s		
Faba bean	c									
Fenugreek			c							
Green bean					p,s					
Green/red gram					p,s					
Hazelnut						c		s		
Kiwi								s		
Lentil	c	s			s			s		
Lima bean	c									
Lipid transfer proteins					s		c	i		
Lupin		s	c	p,c,s	s					
Morintides								s		
Mung bean		s		p						
Pea	c	i,s		s	s					
Peach				i	p			s		
Peanut	s	i,s	c,i	s,p	s,p,i			s,i		
Pomegranate								s		
Red kidney bean			c							
Sesame seed				s						
Soybeans		s	c	s,p	s,c					
Striped catfish									s	
Strawberries					s					
Thaumatin-like proteins							c	i		
Walnut				s				s		
Yellowfish tuna									s	

s, sequence similarities; i, immunological responses; p, phylogenetic relationship; c, cross-reactivity.

(i.e., resistance to digestion) and the ability to elicit IgE-mediated allergenic responses in human and experimental animals. In addition, these allergens (Vig r2–r5) share high sequence similarity with allergens from soybean, lentil, pea, peanut, and lupine, suggesting the potential for cross-reactivity (Misra et al., 2011). Furthermore, allergens Vig r1 (identified in seedlings) and Vig r6 are birch pollen allergen (Bet v1)-homologous and can induce IgE-mediated allergenic cross-reactions in birch pollen allergen-sensitized patients (Guhsl et al., 2014; Mittag et al., 2005). Vig r1 also has high sequence homology with PR-10 proteins from yellow lupine (Koeberl et al., 2018). Additionally, LTPs are also present in mung bean. Studies on phylogenic analyses and sequence homology indicated that mung bean LTPs are closely related to allergenic LTPs from peanuts, peach and green beans and share > 60% homology with LTPs from lentils, beans, peanuts, strawberries and apples (Mishra & Kumar, 2021). Moreover, mung bean proteins produced positive SPTs and immunological responses in peanut-allergic patients and showed cross-reactivity with soybeans (Cho et al., 2017; Jensen et al., 2008). Regarding black gram, multiple IgE-binding

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proteins have been identified; however, to date, the only allergen identified and characterized is the 28 kDa Vig m. This exhibited allergen-like properties (i.e., resistance to digestion) and sequence homology to the rho-specific inhibitor of transcription termination of peanut (Verma et al., 2012, 2013). Furthermore, black gram can cross-react with legumes (i.e., faba bean, lentil, lima bean, and pea) and with the plant *Prosopis juliflora* ("Pollen Food Allergy Syndrome"; Arora et al., 2021; Kumar et al., 2014; Verma et al., 2013).

Apart from legumes, few studies were found for quinoa and hemp seeds. Quinoa exhibited cross-reactivity with hazelnut, which suggests broader cross-reactivities among quinoa and tree nuts (Cho et al., 2017). Concerning hemp seeds, at present five (airway) allergens have been characterized and named as follows: Can s2 (profilin, 14 kDa), Can s3 (nsLTP type 1, 9 kDa), Can s4 (oxygen evolving Enhancer Protein 2, 27.3 kDa), Can s5 (pathogenesis-related protein PR-10, 17.7 kDa), and Can s 7 (thaumatin-like protein [TLP]; WHO/IUIS Allergen Nomenclature Subcommitee). Furthermore, also the protein RuBisCo (Can s RuBisCo, 50 kDa), identified in hemp seeds, can be considered an allergen (Mamone et al., 2019). Most of the allergic reactions reported to hemp are due to inhalation, smoking, and contact associated with the use of *Cannabis* as a drug. Nevertheless, the recent increase in reports of hemp allergy can be attributable to cross-reactivity with inhalation or dietary pan-allergens, that is, ns-LTPs and TLPs. Also, it is speculated that the introduction of hemp-based foods could induce sensitization to secondary plant-derived allergens ("Cannabis Fruit/Vegetable Syndrome"; Mamone et al., 2019).

With regards to leaf proteins, proteins from alfalfa showed IgE-binding inhibition of plant allergens from LTPs, TLPs, and Bet v1-like protein families. Sequence similarities have also been identified with plant allergens, that is, lentils (Len c3), birch pollen (Bet v1), corn (Zea m14) as well as with latex, tree nuts (peanut, hazelnut, and walnut), and fruit (apple, kiwifruit, peach, and pomegranate) allergens, but not with allergens from animal foods. Therefore, potential cross-reactivity only exists for plant-allergic patients (EFSA, 2009; Yakhlef et al., 2021). Additionally, alfalfa can induce indirect histamine release in peanut-allergic subjects (EFSA, 2009; Jensen et al., 2008). Proteins from *M. oleifera* leaves can elicit allergic responses in sensitized mice (Zhang et al., 2022). In silico analyses identified potential *M. oleifera* allergens from Morintides, which share high sequence homology with allergens implicated in fruit-latex syndrome, and with plant pan-allergens ns-LTPs (D'Auria et al., 2023).

Concerning the prevalence of sensitization, this factor varies greatly among countries and is influenced by societal and demographic variables and dietary habits. However, some general patterns have been recognized for legumes. Allergies to lentils (and chickpeas) are most prevalent in the Mediterranean area, especially among children, with lentil sensitization being the most commonly reported. Furthermore, the majority of legume-allergic children have multilegume allergy (> 2), yet lentil is the only legume for which monosensitized patients have been identified (Reche et al., 2013; Soyak Aytekin et al., 2022; Verma et al., 2013). Further, high sensitivity to faba bean is also reported in the Mediterranean area; in detail, adults and children from Morocco exhibited high levels of faba bean allergen sIgE (Bousfiha & Aarab, 2014). As for black gram, it is considered a major allergen in Asian countries, Africa, and India, where it is also commonly consumed (Verma et al., 2012).

Apart from the prevalence, it is also important to consider the effects of food processing on allergenicity as allergens can be thermosensitive and heat treatments can alter protein structure and IgE-binding capacity and eventually reduce the immunoreactivity of foods (Burbano & Cuadrado, 2014; Cuadrado & Pedrosa, 2017). Verma et al. (2013) investigated the processing of lentils and reported that only severe autoclaving eliminated the Len c1 allergen but it had no effect on other immunoreactive proteins. Burbano and Cuadrado (2014) and Cuadrado and Pedrosa (2017) reported that Len c1, as well as Len c2, were isolated from boiled lentils. Overall, consensus holds for the high thermostability of lentil allergens, as boiling alone had little or no effect in reducing the overall allergenicity of lentils. In confirmation, only harsh treatments such as autoclaving and DIC (instant combining pressure drop) can significantly decrease the IgE-binding capacity of lentil allergens, yet extremely resistant immunoproteins may resist such treatments (Burbano & Cuadrado, 2014; Cuadrado & Pedrosa, 2017; Verma et al., 2013). Another treatment effective in reducing the allergenicity of lentils, via the elimination of binding epitopes, is enzymatic sequential hydrolysis (endoprotease Alcalase and exoprotease Flavourzyme); however, some allergenic proteins can resist as they have been detected in the sera of allergic patients after consumption (Cuadrado & Pedrosa, 2017). Furthermore, Skrzypczak et al. (2022) reported that fermentation can also reduce the allergenicity of lentils. For black gram, allergenicity is reduced by heating as Kasera et al. (2012) reported that boiling at 121°C for 15 min reduces sIgE-binding and biopotency of soluble and insoluble protein fractions. Also, the combination of boiling with γ-irradiation further decreases these parameters, whereas, γ-irradiation alone produced no changes in those (Kasera et al., 2012). Verma et al. (2012) further confirmed that black gram allergenicity is time-dependently reduced with boiling, while immunoreactive proteins are not eliminated with roasting (Verma et al., 2013).

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Concerning hemp seeds, hemp allergens do not survive the isolation process for the production of hemp protein isolate (HPI); only short peptides survive gastrointestinal digestion (GID). Therefore, it is believed that HPI can be used in hypoallergenic food formulations (Mamone et al., 2019). Concerning the other protein sources included in the review, no results were retrieved from the searches.

3.2.2 | Aquatic-based alternative protein sources

Concerning duckweeds, these are commonly consumed in Asia and no allergic reactions have been reported there. However, in silico proteomic analyses of whole plants indicated homologies with other plant allergens from celery (Api g3) and with putative airway allergens from hazelnut (Cor a10) and wheat (Tri a31 and Tri a34). Nonetheless, no cross-reactivity has been detected with those foods (Turck et al., 2022). EFSA pointed out that the potential for duckweed to trigger allergic reactions is considered low but not nonexistent owing to the elevated protein content of those plants (Turck et al., 2021b, 2021c, 2022, 2023).

James et al. (2023) recently identified potential allergens of 13, 17, 19, 25–26, 46–50, 72 kDa in microalgae *Chlorella* spp. In silico evaluation of the allergenic potential of *C. vulgaris* extracts has been performed via AllergenOnline (AllergenOnline; search for allergens, and allergen homologous sequences) and Allergome (Allergome, directsearch microalgal proteins) databases and using FAO/WHO constraints for cross-reactivity. Four potential allergens have been identified. The first is calmodulin A0A2P6TFR8; it is related to troponin C proteins that mediate airway allergies and to the Brown Shrimp food allergen Cra c6 (D7F1Q2 protein). However, the latter did not satisfy all the FAO/WHO constraints for cross-reactivity as sequence identity with the calmodulin was smaller than six amino acids. The second protein is fructose-bisphosphate aldolase A0A2P6TDD0, which displayed homologies with allergens or potential allergens from multiple foods. In detail, these are (a) the Pan h3 allergen from striped catfish (XP_026771637 protein), (b) the Thu a3 allergen from yellowfish tuna (D4HTS6 protein), (c) the Sal s3 from Atlantic salmon (B5DGM7 and I0J1J3 proteins), (d) the protein A0A068FCL9, which is related to Chinese white shrimp but is indicated as an allergen in AllergenOnline only. The other two proteins identified were ribosome biogenesis brx1 and cytochrome c-containing protein. However, the first is related to inhalation and skin contact allergenicity while the second is homologous to a filamentous fungus protein that is not considered a food allergen. Overall, the fructose-bisphosphate aldolase protein was recognized as the most likely allergen of *C. vulgaris* because of high sequence homology with edible fish and crustacean allergens (Bianco et al., 2022). Apart from allergen identification, few studies have been retrieved on immunological studies and clinical allergic reactions (Szabo et al., 2013). Szabo et al. (2012) performed a human repeated insult patch test (HRIPT) with Whole Algalin Protein (WAP) and Flour (WAF) from *C. protothecoides*. The experiments revealed no skin sensitization and, hence, such preparations are expected to have little potential to elicit allergic reactions. Recently, for the first time, an anaphylaxis reaction was associated with the consumption of *C. vulgaris* supplements; however, IgE analysis and SPT were not performed on the patient. Yet, after a case of acute tubulointerstitial nephritis, also associated with consumption of *C. vulgaris* supplements, positive SPT and increased nonspecific IgE were detected. Furthermore, positive sIgE and SPT to *C. homosphaera* were identified in mold-sensitive children. However, symptoms of sensitization to *Chlorella* could not be identified in those (James et al., 2023).

4 | DISCUSSION

The results of this systematic literature review indicate that all the emerging alternative protein sources investigated can contribute to some extents to the occurrence of adverse toxicological effects and/or allergic reactions in susceptible individuals.

With respect to the toxicological effects, findings from this review indicated that different threshold values such as the NOAEL, the LD_{50} or the LC_{50} have been established for *Moringa* spp., duckweeds, and microalgae species, as well as their fractions and derived protein ingredients. Additional findings indicated that microalgae grown under stressful conditions can be cytotoxic for mammals and can accumulate toxic compounds from the aquatic web and transfer them to aquatic animals. This ability to absorb and transfer substances is extensively reported in scientific literature, however, a detailed understanding of the modalities (and the risks) is still lacking (Matos, 2017; Milana et al., 2024). Beyond findings on toxicological tests, all the protein sources investigated can produce a variety of secondary metabolites such as ANFs, phytoestrogens and oligosaccharides. At low doses, most of these compounds are considered active ingredients with beneficial effects, such as phytic acid, lectins and phenolic compounds which can reduce blood glucose levels (Popova & Mihaylova, 2019). In contrast, others, such as pyrimidine glycosides, are toxic even at low doses, to people with favism (G6PD-deficient individuals; Ali et al., 2022; Popova & Mihaylova, 2019; Thakur et al., 2019). Overall, it is worth underlining that the occurrences of secondary metabolites

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alone are not sufficient for establishing any level of toxicity or risk. Also, the occurrences reported for the emerging alternative protein sources are not significantly different from those observed in well-established alternative protein sources, such as soybeans (Ali et al., 2022; El-Shemy et al., 2000; Popova & Mihaylova, 2019).

With respect to the allergenic potential, all the protein sources investigated can potentially contribute to the onset of adverse reactions in susceptible individuals in multiple ways. The intrinsic presence of allergens or allergen-like proteins has been reported for all alternative protein sources investigated, except for quinoa and duckweeds. For lentils, mung beans, and hemp seeds, the allergens are also included in the classification system established by the World Health Organization and the IUIS; namely, the WHO/IUIS Allergen Nomenclature Subcommittee (WHO/IUIS Allergen Nomenclature Home Page). To date, the identified allergens for the remaining protein sources are not in that system. This is most likely due to the fact that these foods are still underconsumed in Western diets and therefore perceived as less relevant food allergens, which are consequently less investigated. However, the protein transition requires changes in dietary patterns, so in-depth knowledge is needed to safeguard all consumers. In this connection, none of the plant sources investigated in this study is a regulated and prioritized allergen in Europe or in the United States; which means their (possible) presence does not need to be highlighted on food labels to eventually protect sensitive consumers (European Commission, 2011a, 2011b; FDA, 2004). In Europe, most manufacturing companies use precautionary allergen labeling (PAL) to indicate the presence of regulated allergens in foods. However, a standardized EU approach to the use of PAL is still lacking (Linders et al., 2023). Apart from the presence of allergens, the findings of this review indicate the existence of established or potential cross-reactivities for all protein sources investigated, including quinoa and duckweeds for which intrinsic allergens were not identified. Furthermore, among the different cross-reactivities identified, most occur with the recognized and prioritized allergens such as soybeans or peanuts. In this context, it is worth underlining that the existence of a (potential) cross-reactivity is a useful condition for the identification of allergens, but is by no means sufficient to classify a protein as an allergen. Nevertheless, such knowledge is valuable in protecting consumers as the PAL only addresses regulated allergens, yet allergic consumers may still experience adverse allergic reactions due to cross-reactivity. This is true especially when including multiple alternative plant-based proteins in the diet.

Although the existence of harmful secondary metabolites and allergens in the plant kingdom is widely acknowledged, there are still considerable gaps in conducting a thorough evaluation of the risks associated with the consumption of derived foods. In detail, while it is essential to first understand the presence of these compounds in protein sources, a further step to address their presence in derived foods is crucial to assess the actual risks for consumers. In this context, two main considerations must be made. First, the processing steps applied for the production of protein ingredients and analogues are not clearly established, nor is their impact on phytochemicals and allergens. Second, individual factors such as genetics, lifestyle and consumption patterns play a key role in individual susceptibility (i.e., dose sensitivity); so the actual prevalence of sensitization and the risk levels associated with phytochemicals and allergens are strongly dependent on those external factors (Loh & Tang, 2018; Rennie et al., 2023). However, the last aspect is not easily and unequivocally predictable. Therefore, the evaluation of the toxicological and allergic attributes of emerging alternative protein sources for use in analogues should give more consideration to the first aspect presented, that is, the effect of processing.

In this regard, the lack of standardized production procedures entails that the safety of the final products can vary significantly. Although the primary focus of this review was not the effects of processing, the findings in this regard support common knowledge that the content of ANFs and the allergenicity of foods can both be reduced with processing, especially when multiple strategies are used in combination (Ahmad et al., 2022; Batista et al., 2020; Chen et al., 2021; Dong et al., 2021; Hall & Moraru, 2021a; Vallath et al., 2022). Nevertheless, processing does not always ensure a complete removal. Some findings form this review indicated that ANFs were still present in protein extracts from mung bean and alfalfa. Furthermore, while industrial processing can potentially reduce food safety risks, it can also have negative effects. The processing of plant proteins can affect the functionality and digestibility of protein ingredients, and as well as can lead to the production of harmful processing contaminants, including Maillard compounds and advanced glycation end products (AGEs), compromising food safety (Gräfenhahn & Beyrer, 2024; Lin et al., 2023; Milana et al., 2024; Rivera et al., 2024). The actual effects of processing depend on the conditions applied and the protein source in question. However, there is still limited knowledge and awareness of the undesirable phytochemicals and allergens that could be found in commercial analogues (Gräfenhahn & Beyrer, 2024). Therefore, further research is needed to balance the effective elimination of ANFs and allergens while maintaining functionality and digestibility and minimizing the formation of processing contaminants. Together, these considerations further increase the complexity of performing a comprehensive food safety

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assessment, as neither the safety of analogues nor the level of risk existing for the population can be easily determined. Overall, this also increases the already existing difficulties in protecting the most sensitive consumers.

In light of the above, it is fair to underline that this review by no means intends to discourage the dietary transition or negatively influence public opinion on plant-based foods. In fact, these have been consumed for years in different countries and have a multitude of established health benefits, which were beyond the scope of this paper. As a matter of fact, the aim is to provide a sound foundation to guarantee a preventive approach to food safety and ultimately facilitate this necessary transition.

5 | CONCLUSION

This study applied a systematic literature review to investigate the toxicological effects and allergenic potential of emerging alternative protein sources of plant origin, that is, cowpea, faba beans, lentils, mung beans, black gram, quinoa, hempseeds, leaf proteins, macroalgae, and duckweeds and derived protein ingredients to be used for the production of analogues. The key findings are summarized as follows:

(1) Plant secondary metabolites, such as ANFs and phytoestrogens have been identified in all the protein sources as well as in some derived protein ingredients. In a few cases they have overt adverse health effects and for some protein sources, threshold levels have been established.
(2) All the protein sources investigated can potentially contribute to the onset of adverse allergenic reactions in susceptible individuals. However, none of them are a recognized or prioritized allergen. Intrinsic allergens were found in most protein sources, but only in a few cases these are covered by the allergen nomenclature system. The potential of cross-reactivities was found for all protein sources either with regulated or nonregulated allergens.
(3) Processing treatments generally applied for the production of analogues from protein sources should reduce the risk of experiencing adverse effects with consumption, especially when more techniques are used in combination. However, no studies on the effects of processing on toxicity and allergenicity in commercial analogues are available.

In the perspective of major dietary shifts toward more sustainable diets, the consumption of plant-derived foods will increase significantly, which will result in the increased availability of analogues. Those foods are expected to be increasingly processed to achieve the desired taste, texture, and shelf-life. From a health-related perspective, the increase in consumption of such foods potentially raises dietary exposures to harmful substances such as plant secondary metabolites, allergens, and even compounds developed during processing. In light of the above, the knowledge of such aspects for protein sources and protein ingredients should be combined with the state-of-the-art knowledge from the (ever-evolving) food processing sector so that food safety assessment can be performed at the food level rather than at the ingredient level. This would also facilitate a subsequent step that goes toward an assessment of food safety at the dietary level. In this connection, the alignment among the various stakeholders in the food sector still needs to be maximized to guarantee a safe shifting toward more sustainable diets.

AUTHOR CONTRIBUTIONS

M. Milana: Investigation; writing—original draft; methodology; writing—review and editing; formal analysis; conceptualization. **E. D. van Asselt**: Conceptualization; methodology; writing—review and editing; supervision. **Ine H. J. van der Fels-Klerx**: Conceptualization; methodology; writing—review and editing; supervision.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

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